Peer Effects and Academic Performance: The Impact of Secondary Schoolmates in Undergraduate Classrooms*

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Abstract

This paper exploits random assignment of students to first-semester class groups at a leading Chilean university, combined with data on students' secondary school backgrounds, to examine how exposure to former schoolmates shapes college success. The high concentration of secondary school peers within degree programs, together with random class group assignment, generates exogenous variation in exposure to familiar peers. While peer effects are theoretically ambiguous, our findings show that exposure to familiar peers in the first semester reduces first-year GPA by about 0.04 points (0.08 standard deviations). These effects appear to be driven by increased socialization opportunities that crowd out academic effort during the critical transition period to university life.

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1 Introduction

Peer effects play a crucial role in shaping educational outcomes, yet their mechanisms remain poorly understood, particularly during the critical transition from secondary school to university. This paper examines how pre-existing social connections affect academic performance by studying the impact of sharing first-semester university classes with former secondary schoolmates.

The empirical challenges in studying peer effects are well-documented: peer groups typically form endogenously, and social interactions are complex and difficult to measure. Natural experiments involving random assignment to peer groups thus provide rare opportunities to identify causal peer effects. Sacerdote (2001)'s seminal work on random dormitory assignments sparked a literature exploiting such variation to study various social outcomes. For instance, Boisjoly et al. (2006) examine how random roommate assignment affects attitudes toward other ethnic groups, while Kremer and Levy (2008) study its influence on alcohol consumption. More recently, Corno et al. (2022) leverage random assignment to investigate the role of race in academic interactions.

Our paper builds on this literature but makes a distinct contribution. While previous work has used random assignment to study non-academic interactions (Carrell et al., 2013; Sacerdote, 2001), we combine random class group assignment with detailed data on students' secondary school backgrounds. This unique setting allows us to examine how pre-existing social networks shape academic performance when students transition to university.

Our estimates show that having at least one secondary-school classmate in the first-semester class group lowers first-year GPA by about 0.04 points (0.08 standard deviations). To place this effect size in context, recent studies of random peer assignment in universities typically find positive academic spillovers. For example, Barwick et al. (2025) reports gains of 0.05–0.10 standard deviations in first-year performance when students are assigned to higher-achieving peers. Our setting differs in an important way: peers who share the same secondary school come with strong pre-existing social ties, and our results point to a mechanism in which socialization crowds out academic effort. The comparison highlights that peer effects depend not only on peer ability but also on the nature of prior relationships among peers.

Our analysis uses random assignment of first-year students to class groups at the Pontificia

Universidad Católica de Chile, a leading Chilean university. Two features of our setting are particularly valuable for studying peer effects. First, random assignment addresses the standard identification challenges in peer effects estimation. Second, the high concentration of secondary school graduates within undergraduate programs generates substantial variation in exposure to former peers, a unique feature compared to other university settings where secondary schoolmates rarely share classes.

The effect of secondary schoolmate presence on academic performance is theoretically ambiguous. Familiar peers could ease the transition to university life by providing social support and information sharing. However, they might also increase opportunities for socializing and procrastination, potentially detracting from academic effort.

Barwick et al. (2025) document positive spillovers of approximately 0.05-0.10 standard deviations when students are assigned to academically strong peers. In contrast, we find that exposure to former secondary-school classmates has a negative effect on short-run academic performance: students assigned to a class group with at least one former schoolmate earn first-year GPAs that are about 0.04 points lower, corresponding to roughly 0.08 standard deviations. While the estimated effects on longer-run outcomes such as final GPA, graduation, and academic dismissal are not individually statistically significant in the main specification, the pattern of results is consistent with a socialization mechanism in which pre-existing social ties crowd out academic effort during the critical transition to university.

Our empirical results support the latter mechanism: students assigned to class groups with secondary schoolmates earn lower grades, with the effects concentrated in the first year of study. The findings point to important policy implications for university admissions and student support. The negative effects of secondary schoolmate presence on short-run academic performance suggest potential benefits from strategic class assignments and targeted support during the critical first-semester transition period.

Pre-existing social ties can shape academic outcomes during the transition from secondary school to university. Secondary-school classmates may affect performance through at least two channels. First, strong social connections can increase non-academic interactions that crowd out study time, weaken academic focus, or reduce the incentive to form new academically oriented peer networks. Second, classmates with stronger prior preparation may generate informational or academic spillovers, providing guidance on coursework, study strategies, or navigating uni-

versity requirements. Whether the net effect is positive or negative is therefore an empirical question, and our setting–with random assignment to first-semester class groups–allows us to isolate these mechanisms by comparing students who do and do not share secondary-school peers within the same degree-cohort cell.

The paper proceeds as follows: Section 2 reviews the literature on peer effects in education. Section 3 describes our institutional setting and data. Section 4 details our empirical strategy. Section 5 presents our findings. Section 6 discusses policy implications and concludes.

2 Literature Review

The literature extensively explores the influence of peers on a student's academic performance. Although findings vary among studies, there is a prevailing consensus that peers significantly impact academic outcomes. Peers act as both an external influence and a catalyst, shaping a student's effort and self-confidence. However, this relationship is complex and not straightforward.

Studies utilizing linear-in-means models or fixed effect approaches (Ding and Lehrer, 2007; Vigdor and Nechyba, 2007; Zabel, 2008) show moderate positive impacts of peer ability on academic achievement. Conversely, strong positive peer effects emerge in studies where primary selection is randomized (Duflo et al., 2011; Kang et al., 2007; Whitmore, 2005; Graham, 2008; Zou, 2024). Some studies find peer effects to be evenly distributed (Angrist and Lang, 2004; Arcidiacono et al., 2012), while others observe asymmetric peer effects (Zabel, 2008; Rothstein, 2009; Calvano et al., 2022; Boucher et al., 2024).

Effort patterns vary across different peer environments due to observable and unobservable peer characteristics (Graham, 2008; Epple and Romano, 2011; Kimbrough et al., 2017), adding complexity to this relationship. For instance, empirical evidence suggests that, in tracked settings, students in higher tracks exert substantially more effort than those in lower tracks (Carbonaro, 2005). This positive impact of peers' effort becomes more pronounced when considering the influence of higher-achieving peers on other students through imitation (Cooley, 2009).

The relationship between peers and student outcomes can be even more complex. Zárate (2023) randomly manipulated the characteristics of dormitory neighbors and found that higher-achieving peers can diminish the productivity of lower-achieving students by reducing their

self-confidence. Therefore, even in contexts where higher peer ability directly enhances student performance, the indirect effect on self-confidence can lead to overall negative peer ability spillovers. This suggests that the benefits of higher-achieving peers are nuanced and can have unintended negative consequences for some students.

Furthermore, two other potential key factors, socialization and persistence, have not been exhaustively explored in the peer effects literature. Peers of any ability may indirectly negatively impact student performance, as social activity is often linked to procrastination (Day et al., 2000). Similarly, the presence of peers is associated with increased compulsivity (Gardner and Steinberg, 2005; O'Brien et al., 2011), which positively correlates with procrastination (Gustavson et al., 2014). Conversely, increased socialization with peers may positively impact student performance through channels such as enhanced synergies from additional information exchanges or improved mental health. Additionally, peer effects may persist over time, as documented by Arcidiacono et al. (2012), who report significant and lasting impacts even after relatively brief interactions between students.

The empirical estimation of peer effects is complicated by potential selection problems, given that the formation of peer relationships can be endogenous. An additional complication is the reflection problem, because students can simultaneously affect and be affected by peers (Carrell et al., 2013). Therefore, natural experiments where students are randomly assigned to their peer groups provide a unique opportunity to exploit exogenous group formation. This methodology is introduced by Sacerdote (2001) and Zimmerman (2003), who take advantage of (natural) experiments where students are randomly assigned to peer groups. This methodology has been extensively used to study peer effects on academic performance (Duflo et al. 2011; Garlick 2014; Goulas et al. 2023, among many), as well as on non-academic dimensions and non-classroom interactions (for example, Boisjoly et al. 2006; Kremer and Levy 2008).

Additionally, the relationship between measures of personal characteristics and peer characteristics can result in biased empirical estimates of peer effects. Classical measurement error, usually associated with attenuation bias, can actually lead to substantial overestimation of peer effects, especially in settings where group assignment is systematic (Angrist, 2014). Nevertheless, in contexts with random group assignment, classical measurement error has been shown to cause only typical attenuation bias (Feld and Zölitz, 2017).

We contribute to the empirical peer effects literature by taking advantage of a natural ex-

periment, whereby new undergraduate students were randomly assigned to first-semester class groups, to investigate how the presence of secondary schoolmates impacts academic performance. This idea is similar to Shue (2013), who uses the random assignment of MBA students to class groups to explore how executive peer networks can affect long-term managerial decision making. However, there are two key differences.

First, we focus on the long-term impact of socialization on undergraduate academic performance. Specifically, if sharing the first-semester classroom with peers from the same secondary school increases the likelihood of socialization and expands a student's peer group, a negative (positive) effect of schoolmate presence on academic performance would suggest that the *distractive* (*productive*) effect of socialization prevails in this context. Supporting this idea, Calvano et al. (2022) demonstrate that undergraduate students from the same high school interact more frequently than those without prior connections, making former schoolmates more likely to establish social ties in university settings. These interactions can intensify peer effects, as students may rely on familiar relationships for academic or social support, which can ultimately shape their performance.

Second, in the context of our study it is common for students who had attended the same secondary school to enroll in the same undergraduate degree programs, so that we can observe several secondary schoolmates sharing the same first-semester classroom. This is different than other settings, such as undergraduate degrees at elite universities in the United States, where it would typically be very unlikely to find several secondary schoolmates in the same first-semester classroom.

3 Data

Our empirical analysis uses anonymous administrative data from undergraduate students at PUC, collected routinely for academic and administrative purposes. This includes standard administrative information required for admission (such as secondary education details and standardized test scores) and socioeconomic status information (necessary for determining benefit eligibility). Additionally, once students are enrolled their grades, courses taken, class groups, and academic status are recorded each semester to monitor their progress throughout their undergraduate studies.

Data is available for all students enrolled at the university from 2000 to 2012.[1] To ensure complete records from admission to graduation for all studied cohorts, the analysis is limited to students admitted from March 2000 to March 2006.[2] Similarly, we focus our analysis on students enrolled in the Engineering and Commercial Engineering degrees, which are among the largest in terms of student enrollment,[3] and which did not experience any major curricular changes which could prevent reliable cohort comparisons over time.[4]

During the study period students seeking admission to PUC must take a nationwide standardized test for students aiming to enroll in higher education.[5] This test included two mandatory sections (Spanish and Mathematics) and two optional ones (History, Geography, and Social Sciences, or Science, comprising Biology, Physics, and Chemistry), which together generated a weighted standardized admission score. This was subsequently used in a clearing process at the national level, whereby universities submitted their admissions criteria to a centralized authority, and students ranked their university-degree preferences. The system then cleared in several rounds by allocating students to their highest-ranked preferences based on their scores, and using ranked waiting lists to resolve conflicts.

Admissions to both the Engineering and Commercial Engineering degrees at PUC were highly competitive during the study period, with weighted standardized scores ranging from 705 to 710 points and maximum scores ranging from 825 to 830 points, out of a maximum of 850. The Engineering degree typically took six years to complete, while the Commercial Engineering degree usually took five years. The Commercial Engineering degree in Chilean universities typically encompasses the Economics and Business undergraduate majors offered by universities in the United States.

[1] While administrative records exist prior to 2000, these data were compiled using legacy systems with inconsistent variable definitions and incomplete coverage, making them unsuitable for constructing a consistent panel for our analysis.

[2] Due to the country's location in the Southern Hemisphere, the Chilean academic year starts in March and ends in December.

[3] Smaller population sizes result in limited statistical power, increasing the minimum effect size that can be detected as significantly different from zero. Pooling degrees can enhance statistical power but only allows for the estimation of aggregate peer effects rather than degree-specific ones.

[4] While the Engineering and Commercial Engineering degrees also underwent curricular changes during the study period, these changes were limited enough to allow reliable cohort comparisons.

[5] The standardized admissions test format changed in 2003 from the PAA (Prueba de Aptitud Académica, or *Academic Aptitude Test*) to the PSU (Prueba de Selección Universitaria, or *University Selection Test*). In 2020 the PSU was replaced by the PDT (Prueba de Transición para la Admisión Universitaria, or *Transition Test for University Admission*), which in turn was replaced in 2023 by the PAES (Prueba de Acceso a la Educación Superior, or *Higher Education Admission Test*).

Students may have attended public (government-funded), subsidized (partly government-funded), or private (student-funded) secondary schools. Over the study period the quality of education in subsidized secondary schools was typically lower than in private ones, while the quality of public secondary schools was in turn typically lower than in subsidized ones, with notable exceptions such as the elite Instituto Nacional. While several undergraduate financial aid options were available for students who met socioeconomic and academic criteria, the disparity in the quality of secondary education meant that the majority of students admitted to the most demanded degrees at elite universities in Chile had attended private secondary schools.

Moreover, many students who attended the same secondary school subsequently enrolled in the same university degree program, sometimes even sharing first-semester classrooms with multiple secondary school peers. In our sample, 40.3% of students had at least one former secondary school classmate in their first-semester undergraduate classroom. While some students had up to eight such peers, the vast majority of students had one or two secondary school peers in their classroom, if they had any. Conditional on having at least one secondary school peer in their classroom, 56% had only one, 22% had two, and a further 12% had three.

This clustering of secondary school peers represents a key feature of the natural experiment we exploit in our empirical analysis. Such clustering would be unlikely in other settings, such as undergraduate programs at elite U.S. universities, where first-year students rarely share classrooms with former secondary school classmates. Moreover, in the Chilean context, secondary schools are generally smaller than their U.S. counterparts, with many graduating classes comprising 30-80 students (MINEDUC (2020)). This smaller cohort size makes it more likely that students know each other personally, particularly in private and subsidized schools, which make up the ...majority of our sample (Bellei (2009)).

Table 1 reports the distribution of secondary school types for both degree programs over the study period, and Table 2 reports summary statistics for the estimation sample. The latter includes the main variables used in the analysis: the indicator for having at least one secondary-school classmate in the first-semester college class group, student background characteristics, and academic outcomes.

In the Chilean higher-education system, the concentration of secondary-school peers within elite university programs is a long-standing structural characteristic rather than random varia-

Table 1: Distribution of Secondary School Types Among First-Semester Students

Commercial Engineering	Public	Subsidized	Private
2000	3.3%	11.8%	84.9%
2001	6.5%	4.2%	89.3%
2002	7.5%	10.3%	82.2%
2003	4.4%	8.8%	86.8%
2004	2.3%	7.3%	90.4%
2005	3.2%	7.3%	89.5%
2006	3.3%	5.7%	91.0%
Total	4.4%	8.1%	87.5%
Engineering	Public	Subsidized	Private
2000	6.3%	7.6%	86.1%
2001	8.8%	6.7%	84.5%
2002	6.3%	8.7%	85.0%
2003	8.5%	9.2%	82.2%
2004	12.2%	8.7%	79.0%
2005	7.9%	10.4%	81.7%
2006	11.0%	9.4%	79.6%
Total	8.4%	8.6%	83.0%

Notes: This table reports the percentage distribution of secondary school types among first-semester students in the Commercial Engineering and Engineering degrees from 2000 to 2006. Full detailed counts by group are provided in Appendix Table A.1.

Table 2: Summary Statistics

Variable	Mean	Std. Dev.
1st semester GPA	4.497	0.709
1st year GPA	4.686	0.547
Final GPA	4.955	0.578
Graduated (=1)	0.852	0.355
Dropped out (=1)	0.092	0.290
Dismissed (=1)	0.054	0.226
Has secondary-school classmate in group (=1)	0.403	0.491
Weighted admission score	745.868	25.199
Male (=1)	0.721	0.449
Mother's education (years)	7.933	1.632
Father's education (years)	8.296	1.470
Lives with both parents (=1)	0.786	0.410
From Santiago metropolitan region (=1)	0.812	0.391
Public secondary school (=1)	0.069	0.254
Subsidized secondary school (=1)	0.084	0.277
Observations	3,663	

Notes: Summary statistics are reported for the estimation sample used in the main specification (Table 3), which includes degree-cohort and secondary-school fixed effects. GPA is measured on the Chilean 1-7 scale. "Has secondary-school classmate in group" equals 1 if the student has at least one peer from the same secondary school in their first-semester class group. School-type indicators are defined as mutually exclusive categories.

tion. As shown by Villalobos and Quaresma (2023), a small group of private schools consistently channels students into prestigious degrees-particularly Law, Commercial Engineering, and Civil Engineering-at institutions such as PUC and the University of Chile. This pattern has remained stable over the last decade, with private-school students comprising over 80% of Commercial Engineering and over 70% of Civil Engineering cohorts at PUC. Accordingly, the clustering of peers in our data reflects persistent institutional pathways in elite formation, not episodic or idiosyncratic enrollment patterns.

The Engineering and Commercial Engineering programs are among the largest of PUC's 69 undergraduate programs, and jointly represent a substantial share of the university's student body. In 2012, these two programs enrolled 6,184 students out of a total undergraduate population of 20,868 (29.6%). Their relative weight has remained stable over time: by 2020, enrollment in Engineering and Commercial Engineering rose to 7,412 students out of 27,895 undergraduates (26.6%) according to Pontificia Universidad Católica de Chile (2020). Given this scale and persistence, focusing on these programs provides a comprehensive picture of peer dynamics within the largest and most academically central degrees at the university.

Assignment to class groups in the first semester follows a centralized process: for each degree-by-cohort cell, the Academic Unit assigns students to course sections, and students do not select their own schedules. This mechanism ensures that first-semester class-group allocation is not based on academic or socio-demographic characteristics. Beginning in the second semester, however, the system shifts to autonomous course registration. Students choose courses and sections based on academic priority-determined by cumulative GPA, curricular progress, and approved credits-so the first-semester peer groups do not persist as fixed units. This institutional structure supports our identification strategy by providing random peer exposure in the first semester while allowing for endogenous sorting thereafter.

4 Methodology

To estimate the causal effect of secondary school peers on college academic performance, we exploit random assignment of first-year students to class groups at PUC. First-year students are randomly assigned to class groups (referred to as *sections*), where they take their required core courses together throughout their first semester. These core courses constitute the majority—and in many cases all—of students' first-semester coursework, resulting in substantial shared

time.

The average class group has 49.5 students (SD = 3.6). Given the intensive shared classroom time, many incoming students likely form their primary college social networks during this period. Thus, class group assignment plausibly influences students' social connections throughout their undergraduate years and potentially beyond graduation. The combination of random class group assignment, modest class group sizes, and high concentration of secondary schoolmates within undergraduate programs generates exogenous variation in class group composition. This variation provides a unique opportunity to identify the causal effect of secondary school peer exposure in college classrooms.

Our empirical strategy helps test the socialization hypothesis. If sharing a class group with secondary schoolmates increases social interaction, then negative (positive) effects on academic performance would support the *distractive* (*constructive*) socialization hypothesis. Additionally, finding that first-semester secondary schoolmate exposure affects outcomes beyond the first semester would provide evidence for persistent peer effects.

Our analysis examines several measures of undergraduate academic performance. First, we study academic achievement using three GPA measures: first-semester GPA, first-year GPA, and final undergraduate GPA. Second, we analyze three binary outcomes related to enrollment and completion: graduation (= 1 if the student graduated), voluntary dropout (= 1 if the student voluntarily left their studies), and academic dismissal (= 1 if the student was dismissed for poor academic performance)[6].

We estimate the effect of secondary schoolmate presence in first-semester class groups on these outcomes using degree-cohort and secondary school fixed effects. The degree-cohort fixed effects isolate within-cohort variation in secondary schoolmate presence and account for potential confounders such as curriculum changes or variation in instructional staff. Since randomization occurs within degree-cohort cells, these fixed effects ensure comparisons are made among students subject to the same assignment process.

Students from secondary schools that send more students to the university are mechanically more likely to have secondary schoolmates in their class group. Additionally, secondary schools may influence university performance through differences in academic preparation and peda-

[6]Voluntary dropout and academic dismissal may reflect gaps in academic preparation-particularly for students from public and subsidized secondary schools-or difficulties adapting to the university environment.

gogical approaches. Without appropriate controls, our key independent variable—the presence of secondary schoolmates in the class group—could capture these confounding effects.[7] The secondary school fixed effects help address this concern by controlling for time-invariant characteristics of secondary schools.

Hence, our main specification can be represented by the following linear model:

$$\mathtt{y}_{ijkl} = \beta_0 + \delta_1 \mathtt{m}_{ijkl} + \lambda_l + \eta_k + e_{ijkl} \tag{1}$$

where i indexes students, j indexes class groups, k indexes secondary schools, and l indexes degree-specific university entry cohorts.[8]

The dependent variable $\mathtt{y}_{ijkl}$ represents one of the six academic performance outcomes described above. Our key independent variable, $\mathtt{m}_{ijkl}$, equals one if student i in class group j has at least one secondary schoolmate from school k in their degree program in cohort l.[9] The coefficient of interest, δ_1, captures the effect of secondary schoolmate presence on academic performance in each specification.

To verify random assignment, we examine the distribution of six student characteristics across first-semester class groups: gender (= 1 if male), weighted admission score, mother's education, father's education, housing status (= 1 if living with both parents), and region (= 1 if from Santiago Metropolitan Region). In addition to the group-level balance tests, we also conduct a regression-based randomization check by regressing the leave-me-out count of same-school classmates in the first-semester class group on the full set of baseline student characteristics. The results are reported in Appendix Table A.5. Most estimated coefficients are small and statistically insignificant, and a joint F-test fails to reject the null that all baseline covariates are jointly zero ($F = 1.73$, $p = 0.115$). In both specifications, standard errors are clustered at the class-group level.

[7] Moreover, students more susceptible to peer influence might cluster with former schoolmates, potentially confounding the effect of peer presence with pre-existing motivation differences. Thus, including school-by-cohort fixed effects is a critical part of our identification strategy, as it addresses this concern by controlling for any unobserved characteristics shared by students from the same school and cohort.

[8] As part of our robustness analysis, we also consider a more demanding specification with class-group fixed effects (described in footnote 11).

[9] Results are qualitatively similar when using the count of secondary schoolmates instead of the binary indicator. We focus on the extensive margin due to limited variation in the number of secondary schoolmates per class group.

Tables A.2-A and A.2-B in the appendix present the distribution of these demographic and academic characteristics by class group. Despite the multiple dimensions of student attributes and the relatively modest class group sizes, these tables show that these characteristics are well balanced across groups, and joint orthogonality tests yield large p-values in nearly all cases. This supports the validity of the randomization procedure underlying our identification strategy.

While our baseline empirical strategy relies on the previously discussed degree-cohort and secondary-school fixed effects , we also estimated specifications with more demanding school-by-cohort fixed effects. However, these models are underpowered because many school-cohort cells contain very few observations; the results of this exercise are reported in Appendix Table A.3.

5 Results

Table 3 presents OLS estimates of (1) and reports results from our main specification with degree-cohort and secondary school fixed effects. Table 3 shows that exposure to secondary-school peers reduces short-run GPA, with the effects attenuating for longer-run academic outcomes. The largest estimated impact is on first-semester GPA (-0.052, significant at the 10% level), followed by a smaller but more precisely estimated effect on first-year GPA (-0.042, significant at the 5% level), and a small, statistically insignificant effect on final GPA (-0.014). Outcomes related to dropout, graduation, and academic dismissal show no large or precisely estimated effects. Together, these results indicate that exposure to secondary-school peers affects academic performance primarily through reductions in short-run grades rather than through selection out of the program.

In particular, we note that Table 3, columns (1) and (2) show that the presence of a secondary schoolmate reduces first-semester GPA by 0.05 points (significant at the 10% level) and first-year GPA by 0.04 points (significant at the 5% level). Both estimates represent approximately 0.07 to 0.08 standard deviations of their respective GPA measures[10].

[10]To contextualize this magnitude, GPA scores in Chile range from 1 to 7, with increments of 0.1 and a minimum passing grade of 4.0. Our estimated effects of approximately 0.04–0.05 GPA points may appear small, but they can be consequential for students near the passing threshold. For example, a student with a GPA of 4.0 could fall below the minimum passing mark, potentially leading to course repetition, delayed graduation, or increased risk of dismissal. Therefore, even seemingly modest decreases in GPA carry meaningful academic and practical consequences.

Table 3: Impact on Academic Performance of Presence of Secondary School Mates in First Semester College Class Group

	(1) 1st Semester GPA	(2) 1st Year GPA	(3) Final GPA	(4) Graduation	(5) Drop Out	(6) Dismissal
Secondary School Mate(s) in Group (= 1)	-0.052*	-0.042**	-0.014	-0.005	0.005	-0.002
	(0.031)	(0.021)	(0.022)	(0.013)	(0.011)	(0.008)
R^2	0.17	0.15	0.16	0.15	0.12	0.17
Observations	3,469	3,436	3,469	3,441	3,441	3,441

Notes: This table reports the estimated impact of the presence of secondary school mates in a student's first semester college class group on their academic performance. The sample includes students starting their studies in the Commercial Engineering and Engineering degree programs at one of the leading Chilean universities between 2000 and 2006. The impact of the presence of secondary school mates in the first semester college class group is evaluated on six academic performance measures, one in each column. From columns (1) to (6), the dependent variables are: (1) first semester GPA; (2) first year GPA; (3) final GPA upon graduation; (4) graduation indicator; (5) drop out indicator; and (6) dismissal indicator. This table includes fixed effects for degree–cohort and for secondary school. The summary statistics in Table 2 are computed on the full sample satisfying the baseline restrictions, before singleton observations are dropped by the fixed-effects estimator. The estimation sample in this table is slightly smaller because `reghdfe` drops singleton groups; fewer singletons arise under the secondary-school fixed effects used here than under the more demanding secondary-school-by-cohort fixed effects used in some robustness specifications $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

It might be expected that secondary schoolmate presence could aid the college transition over a longer horizon, potentially offsetting initial academic costs. However, we find no evidence of such reversal. The effect on final GPA (column (3), Table 3) remains negative, though small and statistically insignificant.

A natural question is whether the negative peer effects are driven by the academic ability of former secondary-school classmates. To examine this, we split secondary-school peers into high- and low-achieving groups based on their position in the distribution of weighted admission scores and re-estimate our baseline specifications separately for exposure to high-achieving and low-achieving former classmates. The results are reported in Appendix Table A.4.

We find no evidence that exposure to high-achieving former classmates mitigates the negative effects documented in Table 3. If anything, the estimated effects of having high-achieving secondary-school peers are similar in sign and magnitude to those for low-achieving peers. This suggests that our baseline results are not driven by academic spillovers from high-ability peers but are instead consistent with a socialization mechanism whereby pre-existing social ties crowd out academic effort, regardless of peers' academic ability.

In the appendix, Table A.6 reports estimates from a more demanding specification that includes class-group fixed effects, absorbing all between-group variation within degree-cohort cells and therefore relying on substantially less identifying variation.[11]

As expected, the estimated short-run GPA effects are attenuated relative to Table 3. Nevertheless, the point estimates remain consistently negative, and the effect on final GPA becomes statistically significant under this more demanding specification. In particular, students with a secondary-school classmate in their first-semester class group obtain final GPAs that are about 0.04 points lower (coefficient -0.037, significant at the 10% level), corresponding to approxi-

[11] The specification is given by:

$$\mathtt{y}_{ijkl} = \beta_0 + \delta_1 \mathtt{m}_{ijkl} + \gamma_{jl} + f(N_{kl}) + \sum_{h=1}^{6} \beta_h x_{hijkl} + e_{ijkl}, \tag{2}$$

In (2), N_{kl} denotes the number of students from secondary school k admitted in cohort l, and $f(N_{kl})$ is a set of indicators for the number of students from secondary school k admitted to year l. We include $f(N_{kl})$ proxied by a set of 20 indicator variables capturing the number of students from each secondary school in the same degree–cohort. The mode of this number is zero (26.5%), while the mean and maximum are 4.9 and 23 respectively. In addition, six student-level controls x_{hijkl}, $h \in \{1, \ldots, 6\}$ are introduced. These are: (i) Gender (1 = Male); (ii) Weighted Admission Score; (iii) Mother's Educational Level; (iv) Father's Educational Level; (v) Housing Status (1 = Student lives with both parents); and (vi) Region (1 = Santiago Metropolitan Region).

mately 0.06 standard deviations.

For completion outcomes, neither Table 3 nor Table A.6 yields statistically significant effects, though the point estimates in Table A.6 are larger in magnitude: approximately −2.5 percentage points for graduation and 1.4 percentage points for dismissal, compared to −0.5 and −0.2 percentage points, respectively, in Table 3. Importantly, we find no evidence that secondary-schoolmate exposure affects voluntary dropout: the estimated effect is small (less than one percentage point) and statistically insignificant relative to a sample mean dropout rate of 9%. Taken together, these results indicate that the negative peer effects documented above are not explained by selective attrition, but instead reflect reductions in academic performance consistent with a socialization mechanism that crowds out academic effort.

Additional robustness checks are reported in the Appendix. These include the baseline specification augmented with individual-level controls (Table A.7), a specification with school-by-cohort fixed effects (Table A.8), and subsample analyses by degree (Table A.9), gender (Table A.10), and parental education (Table A.11). Across these exercises, the estimated effects of secondary-school peer exposure remain negative and of similar magnitude to those reported in Table 3, with no specification yielding economically large or sign-reversing estimates. While statistical significance varies across columns due to changes in sample size and identifying variation, the qualitative pattern of results is stable.

In particular, Appendix Table A.7 augments the baseline specification by adding individual-level baseline controls while retaining degree-by-cohort and secondary-school fixed effects. The estimated effect of exposure to secondary-school classmates remains negative and comparable in magnitude to the baseline results in Table 3, providing further evidence that the negative peer effects documented in the main analysis are not driven by observable student characteristics.

Finally Table A.9 reports subsample estimates of the main peer effects, showing that the negative impact of secondary-school peer exposure on GPA is broadly consistent across student subgroups, with no evidence that the results are driven by a particular segment of the sample, and Table A.10 presents additional subsample analyses that confirm the qualitative pattern of the baseline results, indicating that the estimated peer effects are stable across alternative sample restrictions. Table A.11 presents additional subsample analyses splitting the sample by parental education (above vs. below the median of 9 years); the qualitative pattern of negative short-run GPA effects is consistent across both subgroups.

6 Conclusion

This paper exploits random assignment of first-year students to class groups at a leading Chilean university to identify the effects of pre-existing social networks on academic performance. Specifically, we examine how exposure to secondary schoolmates in university class groups influences academic outcomes. Our findings show that students assigned to class groups with secondary schoolmates earn lower grades, particularly in the first year of study.

While we acknowledge the possibility that peer effects in university settings could arise from multiple sources beyond secondary schoolmates, such as roommates, study groups, or orientation peers, several factors support our interpretation that the observed effects are driven specifically by secondary school peers. First, the random assignment of students to first-semester class groups mitigates the risk of endogenous peer selection, as students do not choose their classmates. Second, the first-semester curriculum's structure–where students take most or all of their courses within the same assigned group– intensifies interactions with classroom peers relative to other peer types. Third, the inclusion of degree-cohort fixed effects in our empirical strategy isolates within-cohort variation in secondary schoolmate presence by accounting for potential confounders such as curriculum changes or variations in instructional staff. Similarly, secondary school fixed effects control for time-invariant characteristics of secondary schools.

These findings have several policy implications. First, in settings similar to our study context, universities might improve academic outcomes by separating former secondary schoolmates when assigning first-year class groups. More broadly, if the *distractive* effects of socialization dominate its *productive* effects, institutions should carefully consider policies that facilitate the formation of large peer groups. Second, since the negative effects manifest primarily through lower grades rather than voluntary dropout, universities should prioritize early identification and remediation of academic difficulties.

Prior research suggests that peer effects can vary along key dimensions, such as gender or socioeconomic status. In the Chilean context, heterogeneity may also emerge from differences in school type (public, subsidized, private) or geographic origin (Bellei, 2009). While our sample size may limit statistical power to detect interaction effects precisely, acknowledging these potential sources of heterogeneity is crucial for interpreting the results. Even if a full exploration of these effects is beyond the current scope, recognizing this as a potential mechanism adds

nuance to the findings and opens avenues for future research. Understanding how social distraction mechanisms interact with individual characteristics could enrich the broader literature on peer effects and help refine policy interventions in educational settings.

Table A.1: Distribution of First Semester College Students

		Commercial Engineering Degree				Engineering Degree				
Year	Group	Public	Subsidized	Private	Total	Group	Public	Subsidized	Private	Total
2000	1	1	3	51	55	1	3	2	45	50
	2	4	7	41	52	2	3	2	43	48
	3	1	6	45	52	3	4	1	42	47
	4	1	9	43	53	4	3	7	40	50
	...	...	...	...	...	5	7	4	37	48
	...	...	...	...	...	6	1	3	44	48
	...	...	...	...	...	7	2	5	38	45
	...	...	...	...	...	8	1	5	39	45
	All	7	25	180	212	All	24	29	328	381
2001	1	6	3	45	54	1	5	4	41	50
	2	2	1	50	53	2	3	4	39	46
	3	2	3	49	54	3	3	5	38	46
	4	4	2	47	53	4	5	3	35	43
	...	...	...	...	...	5	7	2	41	50
	...	...	...	...	...	6	2	1	45	48
	All	14	9	191	214	All	25	19	239	283
2002	1	6	3	42	51	1	0	4	44	48
	2	2	12	38	52	2	4	6	39	49
	3	5	4	44	53	3	6	5	35	46
	4	3	3	51	57	4	4	5	40	49
	...	...	...	...	...	5	3	3	39	45
	...	...	...	...	...	6	1	2	47	50
	All	16	22	175	213	All	18	25	244	287
2003	1	4	4	43	51	1	4	3	42	49
	2	0	3	48	51	2	3	8	36	47
	3	1	7	45	53	3	5	4	37	46
	4	4	4	41	49	4	5	6	35	46
	...	...	...	...	...	5	6	3	36	45
	...	...	...	...	...	6	1	2	45	48
	All	9	18	177	204	All	24	26	231	281
2004	1	0	2	52	54	1	6	3	39	48
	2	1	6	49	56	2	9	4	32	45
	3	2	5	43	50	3	7	7	35	49
	4	2	3	53	58	4	4	5	39	48
	...	...	...	...	...	5	5	2	40	47
	...	...	...	...	...	6	4	4	41	49
	All	5	16	197	218	All	35	25	226	286
2005	1	3	3	50	56	1	1	8	41	50
	2	0	5	49	54	2	7	3	35	45
	3	3	6	45	54	3	3	5	41	49
	4	1	2	53	56	4	4	4	34	42
	...	...	...	...	...	5	3	4	40	47
	...	...	...	...	...	6	4	5	37	46
	All	7	16	197	220	All	22	29	228	279
2006	1	0	5	51	56	1	6	2	40	48
	2	2	3	47	52	2	4	6	36	46
	3	3	0	47	50	3	3	6	39	48
	4	2	4	48	54	4	8	6	35	49
	...	...	...	...	...	5	4	3	40	47
	...	...	...	...	...	6	7	4	34	45
	...	...	...	...	...	7	2	5	38	45
	...	...	...	...	...	8	7	3	35	45
	All	7	12	193	212	All	41	35	297	373
Total		123	224	2427	2774		317	326	3142	3785

Notes: Distribution of first semester college students in the Commercial Engineering and Engineering degrees. The data set has been constructed using the administrative data routinely gathered by the university from 2000 to 2012, and it includes all students who entered the Commercial Engineering and Engineering degrees at the university via ordinary admission process between 2000 and 2006.

Note for Tables A.2: The data set has been constructed using the administrative data routinely gathered by the university from 2000 to 2012, and it includes all students who entered the Commercial Engineering degree at the university via ordinary admission process between 2000 and 2006. Assignment of students to their first semester college class group was random. Each cell presents the mean of the balance variable (column) in each class group (row). Balance variables are: (1) weighted admission score, (2) gender (1 = male), (3) mother's educational level, (4) father's educational level, (5) housing status (1 = student lives with both parents), (6) region (1 = Santiago Metropolitan Region), (7) secondary school type (1 = public), (8) secondary school type (1 = subsidized). Reported p-values are for joint orthogonality test across class groups during the same admission year for each of the corresponding balance variables.

Table A.2: Balance by Admission Year and Class Group

A. Commercial Engineering Degree

Year	Group	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	Obs.
2000	1	727.37	0.44	8.24	8.44	0.85	0.84	0.02	0.05	55
	2	726.47	0.58	7.69	8.17	0.77	0.73	0.08	0.13	52
	3	727.61	0.56	8.06	8.44	0.73	0.85	0.02	0.12	52
	4	726.02	0.66	8.19	8.58	0.77	0.83	0.02	0.17	53
	p-value	0.97	0.13	0.23	0.42	0.47	0.40	0.25	0.31	
2001	1	730.52	0.61	7.94	8.20	0.87	0.85	0.11	0.06	54
	2	731.72	0.62	7.77	8.60	0.87	0.92	0.04	0.02	53
	3	732.07	0.44	7.91	8.24	0.74	0.83	0.04	0.06	54
	4	729.24	0.58	7.96	8.23	0.73	0.85	0.08	0.04	52
	p-value	0.84	0.22	0.91	0.43	0.12	0.53	0.35	0.75	
2002	1	734.08	0.55	7.90	8.00	0.90	0.86	0.12	0.06	51
	2	732.70	0.50	8.04	8.17	0.77	0.73	0.04	0.23	52
	3	730.85	0.49	8.11	8.42	0.83	0.87	0.09	0.08	53
	4	731.82	0.40	8.10	8.33	0.83	0.84	0.05	0.05	58
	p-value	0.78	0.38	0.90	0.54	0.36	0.21	0.39	0.01	
2003	1	732.32	0.69	8.08	8.43	0.86	0.88	0.08	0.08	51
	2	732.57	0.39	8.06	8.49	0.78	0.92	0.00	0.06	51
	3	730.76	0.53	7.60	8.19	0.79	0.83	0.02	0.13	50
	4	730.08	0.57	7.96	8.34	0.84	0.86	0.08	0.05	49
	p-value	0.86	0.03	0.47	0.73	0.70	0.56	0.10	0.60	
2004	1	744.45	0.50	8.07	8.38	0.80	0.78	0.02	0.10	54
	2	744.96	0.52	7.89	8.56	0.74	0.86	0.02	0.02	55
	3	744.06	0.49	8.16	8.44	0.82	0.86	0.02	0.06	54
	4	744.96	0.57	7.98	8.50	0.82	0.78	0.07	0.08	50
	p-value	1.00	0.81	0.50	0.78	0.72	0.46	0.09	0.72	
2005	1	741.34	0.50	8.27	8.63	0.86	0.86	0.07	0.08	55
	2	742.86	0.48	7.98	8.44	0.87	0.80	0.02	0.07	52
	3	742.54	0.52	7.89	8.64	0.86	0.83	0.02	0.04	51
	4	746.67	0.62	7.98	8.74	0.81	0.80	0.04	0.27	54
	p-value	0.92	0.81	0.05	0.29	0.46	0.55	0.38	0.22	
2006	1	747.90	0.55	8.33	8.61	0.79	0.87	0.02	0.05	56
	2	746.31	0.48	7.69	8.48	0.89	0.90	0.03	0.07	52
	3	746.25	0.62	7.99	8.72	0.89	0.74	0.01	0.06	54
	4	746.67	0.57	8.08	8.74	0.81	0.86	0.03	0.09	54
	p-value	0.98	0.57	0.20	0.59	0.46	0.55	0.38	0.22	

Table A.2: Balance by Admission Year and Class Group

B. Engineering Degree

Year	Group	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	Obs.
2000	2	739.56	0.86	8.28	8.32	0.70	0.72	0.06	0.04	50
	3	738.34	0.85	7.85	8.13	0.75	0.75	0.06	0.04	48
	4	740.70	0.83	8.26	8.38	0.74	0.85	0.09	0.02	47
	5	739.35	0.80	7.98	8.18	0.84	0.84	0.06	0.14	50
	6	739.01	0.81	8.17	8.50	0.79	0.81	0.15	0.08	48
	7	739.82	0.79	7.69	8.06	0.85	0.92	0.02	0.06	48
	8	738.85	0.76	7.78	8.11	0.78	0.80	0.04	0.11	45
	9	739.41	0.87	7.73	8.24	0.80	0.80	0.02	0.11	45
	p-value	1.00	0.86	0.40	0.83	0.64	0.32	0.26	0.30	
2001	1	753.33	0.92	7.72	7.82	0.86	0.80	0.10	0.08	50
	2	752.57	0.85	7.43	8.39	0.85	0.85	0.07	0.09	46
	3	751.14	0.83	8.07	8.26	0.78	0.85	0.07	0.11	46
	4	749.02	0.91	7.77	8.40	0.86	0.84	0.12	0.07	43
	5	750.31	0.88	7.52	7.84	0.86	0.86	0.14	0.04	50
	7	749.99	0.83	8.08	8.54	0.85	0.79	0.04	0.02	48
	p-value	0.92	0.67	0.32	0.07	0.90	0.93	0.54	0.57	
2002	1	738.03	0.90	7.63	8.13	0.77	0.73	0.00	0.08	48
	2	738.72	0.92	8.04	8.29	0.82	0.84	0.08	0.12	49
	3	738.20	0.87	7.87	8.24	0.76	0.72	0.13	0.11	46
	4	737.26	0.88	7.71	7.96	0.78	0.84	0.08	0.10	49
	5	740.09	0.80	7.69	8.53	0.73	0.78	0.07	0.07	45
	7	737.96	0.78	8.18	8.34	0.84	0.78	0.02	0.04	50
	p-value	0.99	0.32	0.49	0.51	0.84	0.61	0.11	0.73	
2003	1	738.23	0.88	7.84	8.16	0.76	0.82	0.08	0.06	49
	2	739.12	0.79	7.89	8.00	0.70	0.74	0.06	0.17	47
	3	736.76	0.65	8.00	8.30	0.78	0.80	0.11	0.09	46
	4	738.45	0.89	7.33	7.67	0.70	0.67	0.11	0.13	46
	5	738.15	0.87	7.53	8.36	0.76	0.64	0.13	0.07	45
	7	736.55	0.79	8.38	8.56	0.71	0.73	0.02	0.04	48
	p-value	0.99	0.03	0.08	0.24	0.91	0.35	0.44	0.26	
2004	1	757.16	0.88	8.00	8.52	0.83	0.85	0.13	0.06	48
	2	756.79	0.98	7.31	7.73	0.73	0.80	0.20	0.09	45
	3	757.44	0.80	7.67	8.31	0.80	0.76	0.14	0.14	49
	4	755.68	0.90	8.06	8.06	0.75	0.77	0.08	0.10	48
	5	757.35	0.83	8.06	8.28	0.79	0.77	0.11	0.04	47
	7	756.69	0.84	7.96	8.04	0.71	0.80	0.08	0.08	49
	p-value	1.00	0.14	0.24	0.24	0.76	0.87	0.52	0.61	
2005	1	767.66	0.92	7.92	7.96	0.76	0.84	0.02	0.16	50
	2	768.24	0.96	7.89	8.27	0.69	0.80	0.16	0.07	45
	3	766.14	0.88	7.61	7.82	0.80	0.78	0.06	0.10	49
	4	767.75	0.81	7.93	8.31	0.81	0.76	0.10	0.10	42
	5	765.61	0.74	7.87	8.26	0.91	0.74	0.06	0.09	47
	7	767.03	0.89	7.91	8.17	0.76	0.80	0.09	0.11	46
	p-value	0.99	0.04	0.94	0.64	0.17	0.90	0.26	0.77	
2006	1	774.80	0.85	8.10	8.38	0.67	0.85	0.13	0.04	48
	2	774.96	0.74	8.24	8.35	0.70	0.85	0.09	0.13	46
	3	772.46	0.79	7.88	8.58	0.83	0.77	0.06	0.13	48
	4	774.47	0.86	7.61	7.87	0.80	0.76	0.16	0.12	49
	5	774.14	0.83	8.00	8.17	0.79	0.79	0.09	0.06	47
	6	772.69	0.82	8.00	8.24	0.71	0.73	0.16	0.09	45
	7	773.30	0.84	8.07	8.24	0.73	0.73	0.04	0.11	45
	8	774.16	0.93	8.18	8.38	0.71	0.82	0.16	0.07	45
	p-value	1.00	0.41	0.66	0.89	0.57	0.72	0.43	0.74	

Table A.3: Peer Effects with Secondary-school-by-cohort Fixed Effects

	(1) 1st Sem GPA	(2) 1st Year GPA	(3) Final GPA	(4) Graduation	(5) Drop Out	(6) Dismissal
Average share of same-school classmates	0.080	-0.139	-0.140	0.005	-0.046	-0.040
	(0.534)	(0.412)	(0.436)	(0.231)	(0.248)	(0.181)
Observations	2992	2963	2992	2970	2970	2970

Notes: Standard errors in parentheses. Fixed effects: degree-by-cohort and secondary-school-by-cohort. Baseline controls included. Standard errors clustered at the class-group–degree–cohort level.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.4: Heterogeneity by Peer Ability

Table A.4 examines whether the effect of secondary-schoolmate presence differs by peer academic ability. We classify peers above the 80th percentile of the weighted admission score distribution as high-ability, and those below the 20th percentile as low-ability. Interacting these indicators with the treatment yields coefficients with signs consistent with the baseline but imprecisely estimated due to limited subgroup size. The results suggest that the negative short-run effects are not driven by exposure to unusually strong or weak peers but reflect mechanisms common across peer-ability levels.

Table A.4: Heterogeneity by Peer Ability

	Outcomes			
	1st Sem GPA	1st Year GPA	Final GPA	Graduation
High	-0.069	-0.036	-0.007	-0.007
	(0.045)	(0.040)	(0.036)	(0.024)
Low	-0.024	-0.008	-0.016	0.005
	(0.037)	(0.024)	(0.031)	(0.021)
R^2	0.38	0.39	0.32	0.17
N	3,469	3,436	3,469	3,441

Notes: Low ability defined as bottom 20% of the ability distribution. All regressions include degree–cohort and secondary-school fixed effects. Standard errors are clustered at the class-group level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.5: Balance Test

This table reports a formal balance test using a leave-me-out measure of peer exposure, showing that baseline characteristics do not jointly predict peer exposure once school and degree-cohort fixed effects are included.

Table A.5: Formal Balance Test: Predictors of Leave-Me-Out Same-School Peer Exposure

	Dependent variable: Leave-me-out count of same-school classmates		
	Coef.	Std. Err.	p-value
Weighted admission score (b_score2)	−0.00059	0.00091	0.520
Male ($b_sex = 1$)	−0.07979	0.04575	0.084
Mother's education (b_educm)	−0.01255	0.01139	0.274
Father's education (b_educf)	0.02725	0.01317	0.042
Housing background (b_house)	0.05958	0.04197	0.160
Region (b_region)	−0.00312	0.20260	0.988
Subsidized secondary school ($b_colegio_subv$)	0.67824	0.57465	0.241
Constant	1.06311	0.72123	0.144
Joint F-test of controls		$F = 1.73$, $p = 0.115$	
Observations		3,473	

Notes: The dependent variable is the leave-me-out count of classmates from the same secondary school in the first-semester class group. All specifications include degree-by-cohort and secondary-school fixed effects. Standard errors are clustered at the class-group-by-degree-by-cohort level.

Table A.6: Peer Effects with Class-Group Fixed Effects

This table reports estimates of the effect of having at least one secondary-school classmate in the first-semester college class group using a specification that includes class-group fixed effects and baseline controls. Class-group fixed effects absorb all between-group variation within each degree–cohort cell, so identification comes exclusively from within–class-group variation in secondary-school peer exposure. To address the fact that students from larger secondary schools are mechanically more likely to have former schoolmates in the same class group, we additionally control for the number of students from each secondary school admitted in the same degree–cohort using a set of indicator variables.

Table A.6: Peer Effects with Class-Group Fixed Effects (Within Class-Group Variation)

	GPA Outcomes			Completion Outcomes		
	1st Sem GPA	1st Year GPA	Final GPA	Graduation	Dropout	Dismissal
Same-school peer in group (=1)	-0.015	-0.026	-0.037*	-0.025	0.008	0.014
	(0.026)	(0.019)	(0.019)	(0.015)	(0.012)	(0.009)
R^2	0.31	0.30	0.23	0.07	0.04	0.07
N	3,660	3,625	3,660	3,634	3,634	3,634

Notes: All regressions include fixed effects for class group and admitted secondary schoolmates, along with baseline controls. Standard errors are clustered at the class group level. Compared to the baseline specification in Table 3, this specification is substantially more demanding and relies on less identifying variation; as a result, estimates are expected to be less precise. The sample and outcome definitions are the same as in Table 3. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.7: Impact of Secondary-School Peers with Baseline Controls

Table A.7 augments the baseline specification (Table 3) by adding individual-level baseline controls while retaining degree-by-cohort and secondary-school fixed effects. The estimated effects of exposure to secondary-school peers remain similar in sign and magnitude, indicating that the main findings are robust to the inclusion of observable student characteristics.

Table A.7: Peer Effects with Baseline Controls

	GPA Outcomes			Completion Outcomes		
	1st Sem GPA	1st Year GPA	Final GPA	Graduation	Dropout	Dismissal
Same-school peer in group (=1)	-0.041	-0.034**	-0.007	-0.003	0.004	-0.003
	(0.025)	(0.016)	(0.020)	(0.013)	(0.011)	(0.008)
R^2	0.38	0.39	0.32	0.17	0.13	0.19
N	3,469	3,436	3,469	3,441	3,441	3,441

Notes: This table augments the baseline specification (equation 1) with individual-level controls (admission score, gender, parental education, housing, region, school type). All regressions include degree–cohort and secondary-school fixed effects. Standard errors clustered at the class-group-by-degree-by-cohort level. GPA measured on Chilean 1–7 scale. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.8: Secondary-School Peers with School-by-Cohort Fixed Effects

Table A.8 reports a specification using secondary-school-by-cohort fixed effects in place of the baseline time-invariant secondary school fixed effects, while continuing to include degree-by-cohort fixed effects. No baseline controls are included. The results remain qualitatively similar to those in the main specification, indicating that the estimated impact of exposure to secondary-school peers is robust to this more demanding fixed-effects structure.

Table A.8: Peer Effects with School-by-Cohort Fixed Effects

	1st Sem GPA	1st Year GPA	Final GPA	Graduation	Drop Out	Dismissal
Same-school peer in group (=1)	-0.028	-0.028	-0.009	-0.003	-0.003	0.002
	(0.034)	(0.023)	(0.023)	(0.014)	(0.013)	(0.008)
R^2	0.28	0.26	0.27	0.26	0.25	0.26
N	2,992	2,963	2,992	2,970	2,970	2,970
Fixed Effects	Degree-Cohort + Secondary-School-Cohort					

Notes: This table replaces the baseline secondary-school fixed effects with secondary-school-by-cohort fixed effects; no individual-level baseline controls are included. All regressions include degree–cohort fixed effects. Standard errors clustered at the class-group-by-degree-by-cohort level. GPA measured on Chilean 1–7 scale. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.9: Subsample Analyses by Degree

As a further robustness check, we estimate the baseline specification separately for Engineering and Commercial Engineering students. Although precision declines due to smaller subsample sizes, the sign and pattern of coefficients remain consistent across both degrees.

Table A.9: Impact of Presence of Secondary School Mates – Subsample Analyses by Degrees

	GPA Outcomes			Completion Outcomes		
	1st Sem GPA	1st Year GPA	Final GPA	Graduation	Dropout	Dismissal
Engineering Only						
Secondary school mate(s) in group (=1)	-0.057	-0.054*	-0.039	-0.022	0.012	0.008
	(0.043)	(0.031)	(0.031)	(0.018)	(0.017)	(0.009)
R^2	0.19	0.18	0.17	0.17	0.16	0.16
N	2,003	1,992	2,003	1,995	1,995	1,995
Commercial Engineering Only						
Secondary school mate(s) in group (=1)	-0.096*	-0.060*	-0.018	-0.007	0.010	-0.004
	(0.053)	(0.033)	(0.041)	(0.021)	(0.017)	(0.017)
R^2	0.20	0.17	0.18	0.23	0.19	0.23
N	1,335	1,312	1,335	1,321	1,321	1,321

Notes: Baseline specification (equation 1). Engineering and Commercial Engineering samples estimated separately. All regressions include degree–cohort and secondary-school fixed effects. Standard errors clustered at the class-group-by-degree-by-cohort level. GPA measured on Chilean 1–7 scale. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.10: Subsample Analyses by Gender

We further estimate the baseline specification separately by gender. Similarly to the degree separation, the sign and pattern of coefficients remain consistent across both genders.

Table A.10: Impact of Presence of Secondary School Mates – Subsample Analyses by Gender

	GPA Outcomes			Completion Outcomes		
	1st Sem GPA	1st Year GPA	Final GPA	Graduation	Dropout	Dismissal
Male Students Only						
Secondary school mate(s) in group (=1)	-0.047	-0.046*	-0.015	-0.021	0.016	0.002
	(0.036)	(0.024)	(0.028)	(0.018)	(0.014)	(0.011)
R^2	0.17	0.17	0.18	0.16	0.13	0.19
N	2,491	2,469	2,491	2,469	2,469	2,469
Female Students Only						
Secondary school mate(s) in group (=1)	-0.096**	-0.043	-0.032	0.022	-0.009	-0.013
	(0.047)	(0.036)	(0.033)	(0.028)	(0.028)	(0.011)
R^2	0.26	0.21	0.19	0.19	0.16	0.30
N	912	901	912	906	906	906

Notes: Baseline specification (equation 1). Male and female student samples estimated separately. All regressions include degree–cohort and secondary-school fixed effects. Standard errors clustered at the class-group-by-degree-by-cohort level. GPA measured on Chilean 1–7 scale. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.11: Subsample Analyses by Parental Education

We further estimate the baseline specification separately by parental education, splitting the sample at the median of average parental education (9 years). The sign and pattern of coefficients remain consistent across both subgroups.

Table A.11: Impact of Presence of Secondary School Mates – Subsample Analyses by Parental Education

	GPA Outcomes			Completion Outcomes		
	1st Sem GPA	1st Year GPA	Final GPA	Graduation	Dropout	Dismissal
High Parental Education (Average $\geqslant$ 9 Years)						
Secondary school mate(s) in group (=1)	-0.075**	-0.051*	-0.028	-0.011	0.004	0.005
	(0.037)	(0.026)	(0.027)	(0.020)	(0.015)	(0.010)
R^2	0.16	0.14	0.14	0.16	0.15	0.17
N	1,776	1,764	1,776	1,766	1,766	1,766
Low Parental Education (Average < 9 Years)						
Secondary school mate(s) in group (=1)	-0.061	-0.056	-0.033	-0.014	0.010	0.001
	(0.056)	(0.038)	(0.037)	(0.025)	(0.023)	(0.017)
R^2	0.24	0.21	0.20	0.21	0.19	0.22
N	1,596	1,576	1,596	1,578	1,578	1,578

Notes: Baseline specification (equation 1). High and low parental education subsamples estimated separately, split at the median of average parental education (9 years). All regressions include degree–cohort and secondary-school fixed effects. Standard errors clustered at the class-group-by-degree-by-cohort level. GPA measured on Chilean 1–7 scale. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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